

Lecture 13: Catch-up

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- 1 Where we are with our running examples- Randomized Kaczmarz (L3): a distribution $p_i \propto \|a_i\|^2$ built from data; error decreases geometrically *in expectation*.- Randomized trace estimation (L4, L10, L11): an unbiased estimator via linearity; variance shrinks like $1/m$ when averaging; Rademacher MGF is sub-Gaussian.- Randomized matrix multiplication (L9): another unbiased, derived-distribution estimator, same shape as trace estimation.. . .::: {.callout-warning icon=false}### FactAll three are *unbiased randomized estimators* – the tools we’ve built (linearity, variance, MGFs) apply to all of them. That’s the whole point of running examples.:::—## Side-by-side: two unbiased estimators

```
n = 15A = rng.normal(size=(n, n))true_trace = np.trace(A)zs = rng.choice([-1, 1], size=(3000, n))trace_
```

Any questions from Homeworks 1-6 or the midterm are fair game today.## Reading & Homework*Reading: review (Bertsekas & Tsitsiklis, Introduction to Probability).*